

Chandana Anand Rangappa

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EDUCATION

University of California, Riverside

Master of Science, Computer Engineering

Riverside, CA

Sep. 2024 – Mar. 2026

- **Coursework:** Distributed Systems, Algorithms, Big Data, ML, AI, Database Systems, Information Retrieval
- **Teaching Assistant, SOC 135: Social Conflict** Graded assignments and exams for **80+** students.

Dayananda Sagar College of Engineering

Bachelor of Engineering, Computer Science & Engineering

Bengaluru, India

Sep. 2020 – Jun. 2024

- **Coursework:** Data Structures, Algorithms, Operating Systems, Database Management, Computer Networks, OOP

EXPERIENCE

Consultant

Capco

Jul. 2026 – Present

Remote, United States

- Built a **RAG** pipeline using **Python**, **LangChain**, **GPT-4**, and **Elasticsearch** embeddings to match flagged bank communications against **SEC** enforcement cases and index past cases for instant lookup, cutting review time **~45%**.
- Designing a verification stage that reranks and cross-checks every AI flag against its source, raising measured accuracy to **~80%**, and deploying it as an internal **Palantir Foundry** web app with usage logging.
- Managing the multi-strategy expansion of **Deal Close** on **Palantir Foundry** across the full deal lifecycle, running sprints and **6 releases** and bridging developers and business users.
- Gathering requirements from Deal Ops, Legal, Compliance, and IR, documenting them as **BRDs** and **user stories**, and presenting AI findings to compliance and legal stakeholders in plain terms.

Graduate Researcher

University of California, Riverside

Jan. 2026 – Jun. 2026

Riverside, CA

- Built an end-to-end decompilation recovery pipeline using **Ghidra**, **Joern**, and **GPT-OSS 120B** to reconstruct structural properties lost during binary decompilation, achieving **97.6%** structural similarity against ground truth.
- Improved structural recovery from **67.2%** to **98.6%** by extracting struct definitions and control-flow patterns via **Joern** CPG analysis and injecting them as structured **RAG** context into the **LLM**, cutting hallucinations across **100k+** programs.
- Designed a worker-coach agent architecture with **ChromaDB**, embedding programs with a **Hugging Face** model to retrieve relevant context, reducing pipeline runtime by **40%**.

Software Engineering Intern

Anand Diagnostic Laboratory

Dec. 2023 – May 2024

Bengaluru, India

- Replaced subjective manual zone grading with an automated **computer vision** system: annotated plate images with **JSON** bounding boxes and trained **YOLOv5** with **OpenCV** and **TensorFlow** for objective inhibition zone detection at **95%** accuracy.
- Built a full-stack web app (**React**, **Flask**, **REST APIs**) with **MySQL**-backed storage, user auth, and a live results dashboard, processing **100+** samples/day and cutting lab reporting time by **30%** through automated reporting.
- Containerized with **Docker** on **AWS (EC2, S3)**, set up **GitHub Actions CI/CD** for weekly deployments, wrote **pytest** tests achieving **~80%** coverage, and secured routes with **OAuth** auth and password hashing over **SSH**.

PROJECTS

AI-Assisted Semantic Property Discovery | Claude API, Groq API, MCP, Hypothesis, Python

- Built an agentic **LLM** pipeline using **Claude** and **Groq** APIs with **MCP** integrations to extract semantic properties and auto-generate **Hypothesis** test cases, catching **60%** of bugs across **250 programs**.
- Evaluated **LLM** output across **250** programs, identifying hallucination and over-generalization as dominant failure patterns.

U-ASK: Spatial-Textual Search Engine | Python, PostgreSQL, Elasticsearch, PostGIS

- Built a search engine that finds places by keyword and distance together, combining a quadtree spatial index with an inverted text index over **5M+** geotagged records at sub-**50ms** latency (**3,500 QPS**).
- Sped up search **30x** over a brute-force baseline by pre-filtering candidates with **PostGIS** before ranking them in **Elasticsearch**.

TECHNICAL SKILLS

Programming & Tools: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS, Git, Linux

Frameworks & Libraries: React.js, Flask, Node.js, REST APIs, PyTorch, TensorFlow, Apache Spark, PySpark, NumPy, Pandas

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Docker, Kubernetes, Elasticsearch, Kafka

AI & ML: LLMs, Agentic AI, RAG, Claude API, Groq API, Hugging Face, BERT, FAISS, scikit-learn, Computer Vision, NLP

Software Engineering: Distributed Systems, System Design, Microservices, CI/CD, Agile, OOP

PUBLICATIONS & RECOGNITION

Publications: **Inhibition Zones: A Window into Antibiotic Resistance**, ML pipeline for antibiotic testing

IoT Solutions for Food Quality Monitoring, ESP32 and sensor-based real-time food safety monitoring system

Recognition: Won Best Innovative Project Award (Nokia Open Day, DSCE) for the Antibiotic Susceptibility Testing System Subcommittee Chair, Student Services Fee Advisory Committee – UC Riverside